Handwritten Notes & Further Reading — Prof. Aaronson Distinguished Lecture

Talk title	Computational Complexity and Explanations in Physics
Speaker	Prof. Scott Aaronson (University of Texas at Austin)
Quarter/year talk was given	Winter 2026 (delivered Thursday, January 8, 2026, 3:30–4:30 PM)
Talk URL/link	https://www.youtube.com/watch?v=GcbLL4C3VwY
Talk was a	Distinguished Lecture
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1. Framing & Computational Complexity

The talk was originally designed for a philosophy audience.

Computational complexity is the study of the inherent resources (time, memory, etc.) needed to solve computational problems, and how those resources scale with problem size. Other resources besides time and memory include **randomness** and **quantumness**.

Scaling behavior — whether a problem gets easier or harder as input grows — is the central concern. Sources of **computational intractability** in practice:

- Limited information
- Chaos (sensitive dependence on initial conditions)
- Irrationality / bias

Historical arc: from Euclid's GCD algorithm (~300 BCE) → the 1930s formalization of computability → Alan Turing and the imitation game → modern large-scale compute (LHC clusters for analyzing particle collisions, and today roughly half the world's economic output going into AI buildout for doing science/physics).

A key thought-experiment about chaos: even an infinitely fast computer wouldn't let you predict weather perfectly — you'd need to model every atom.

Prof. Aaronson's own reference: his paper **"Why Philosophers Should Care About Computational Complexity"** (arXiv:1108.1791).

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2. Three Case Studies

Prof. Aaronson organizes the talk around three case studies:

1. Quantum computing and the Many-Worlds interpretation
2. Complexity-theoretic Occam's Razor
3. Daniel Harlow & Patrick Hayden — regime of validity of QFT / Relativity; what happens if P = NP

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3. Case Study 1 — Quantum Computing & Many-Worlds

Two framing questions:

1. How do quantum computers get such huge speedups?
2. Does that force us to posit the reality of parallel universes?

3.1 Quantum mechanics as "probability theory with minus signs"

Double-slit experiment:

- Classical probability intuition: $P_1 = P_2 = \frac{1}{2}$ each slit; with both open, $P = 1$.
- Quantum reality: amplitudes can be **negative** (or complex), and the observed probability is $P = |\alpha|^2$.
- Example: $\alpha_1 = \frac{1}{2}$, $\alpha_2 = -\frac{1}{2} \;\Rightarrow\; \alpha = \alpha_1 + \alpha_2 = 0 \;\Rightarrow\; P = |\alpha|^2 = 0$ (destructive interference).
- By *decreasing* the number of paths leading to a spot, you can *increase* the probability at another spot.

  "God prefers complex numbers and the 2-norm over the 1-norm."

3.2 Quantum state formalism

A quantum state over n qubits is a superposition:

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle$$

where the α_x are complex numbers called **amplitudes**, and the $|x\rangle$ are orthogonal basis vectors in a 2^n-dimensional Hilbert space.

- A **qubit** is the basic building block — a bit in superposition of $\{0, 1\}$.
- Superposition: before measurement, the system is a linear combination of basis vectors.
- For 2 qubits you need 4 amplitudes; for n qubits you need 2^n complex amplitudes.
- For ~1000 particles, you'd need 2^{1000} amplitudes — **a staggering amount for nature to track**.

3.3 Born Rule & measurement

- **Born rule:** probability of observing basis state x is $|\alpha_x|^2$.
- You never directly "see" the amplitudes. Classical computers are overwhelmed by the size and can't store or manipulate them.
- When you measure n qubits, you force nature to make a decision about which outcome to give you — you get one classical bit-string, with probabilities set by the amplitudes.

3.4 Main known applications

1. **Breaking current public-key cryptography** (Shor's algorithm) — factoring, number theory, group theory, discrete log, elliptic curve. Named after Peter Shor. Motivates a full security-infrastructure upgrade to **quantum-resistant cryptography** (lattice problems). Roughly **$200 billion in Bitcoin** is locked on elliptic-curve keys that Shor would break.
2. **Simulating quantum physics** — the natural and most defensible application. **Grover's algorithm** gives a square-root speedup for unstructured search ($\sqrt{N}$), not exponential. Every algorithm can be "Groverized," but the speedup is only quadratic.

3. **Everything else** — AI, optimization. Ewin Tang's PhD thesis on **dequantization**: many proposed quantum ML advantages can be matched classically, so the claimed quantum breakthroughs in ML often aren't breakthroughs at all.

3.5 Practical thresholds

- Quantum error correction currently costs ~10^6 overhead to protect 1 logical qubit.
- So for Grover to beat classical: $\sqrt{n} \cdot 10^6 > n$ fails until roughly $n > 10^{12}$.
- Translation: the problem must be **larger than ~10^{12} items** before Grover is even worth running.
- Ewin Tang — UW grad student, now Princeton — dequantizing quantum ML.

3.6 Complexity class landscape

- **P** — problems solvable in polynomial time on a classical computer (linear equations, matching, etc.).
- **NP** — problems where a classical computer can efficiently *verify* an answer (graph isomorphism, post-quantum crypto, factoring, discrete log).
- **NP-complete** — the hardest problems in NP (3SAT, packing).
- **BQP** — Bounded-Error Quantum Polynomial Time. Overlaps NP but isn't contained in it; contains factoring and discrete log. For many problems in BQP, a classical computer can't even *check* the quantum answer.
- Famous open problem: whether P = NP.

3.7 Quantum supremacy

- Google's 103-qubit superconducting demonstration — classical computer would take ~10^85 years to simulate it directly.
- Fermi-Hubbard model, Out-of-Time-Order Correlators (OTOCs) are candidate comparison benchmarks.
- Verification is the hard part: you either cross-check with a second quantum computer or with a physical experiment.

3.8 Many-Worlds Interpretation (Everett, 1957)

Measurement of state $|\psi\rangle$ is just us and our environment getting entangled with $|\psi\rangle$, as the world splits into equally-real branches:

$$(\alpha|0\rangle + \beta|1\rangle)\,|\text{you}\rangle \;\longrightarrow\; \alpha|0\rangle|\text{you}_0\rangle + \beta|1\rangle|\text{you}_1\rangle$$

Niels Bohr hated it; when asked to censor it he went on to become a nuclear-weapons strategist.

Main alternatives to Many-Worlds:

- **Bohm / de Broglie pilot wave** — one "real" branch with giant amplitude, the rest are ghosts.
- **New physics** (e.g. dynamical collapse mechanisms).
- **Copenhagen** — "learn to stop asking the question."

3.9 David Deutsch's 1997 challenge

> "When Shor's algorithm has factored a number using ~10^{500} times the computational resources that can be seen to be present, where was the number factored? There are only 10^{80} atoms in the entire visible universe."

Many-Worlds answer: in the other branches. "Is anybody home in the other branches?" Quantum computing itself is silent on this.

Two worlds colliding — that's a quantum computer.

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4. Case Study 2 — Complexity-Theoretic Occam's Razor

Thought experiment: **Zeno's finite-time Goldbach decider.** If you could keep halving the time per step, you could check Goldbach's conjecture in finite time.

> Goldbach's conjecture: every even number 4 or greater can be written as the sum of two primes (4 = 2 + 2). Unsolved.

Why can't you actually do this?

- Lower-limit 1: **Planck time** $\approx 10^{-43}$ s. Go any faster per step and the clock cycle concentrates enough energy in a small region to exceed the Schwarzschild radius — **your computer collapses into a black hole.**
- **Relativity computing:** the energy needed to accelerate to c diverges, and there's a tradeoff where time saved is exponential in energy spent.

Prof. Aaronson references recent Santa Fe Institute work with **Gulce Kardes and Harrison Hartle** on this.

Core insight: if $P = NP$, you could appear to reverse the **Second Law of Thermodynamics** — build a Maxwell's demon. Physics and computational complexity are propping each other up.

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5. Case Study 3 — Harlow & Hayden

Daniel Harlow and **Patrick Hayden** — the regime of validity of quantum field theory and relativity, viewed through complexity.

If $P = NP$, Maxwell's demon becomes buildable → breaks thermodynamics.

Harlow–Hayden's black-hole-information argument: decoding Hawking radiation to recover infalling information requires a computation roughly of magnitude $2^{10^{67}}$ — doubly exponential. The firewall paradox is dissolved because the adversary that would produce the paradox literally cannot exist by computational-complexity grounds.

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6. Closing themes

- There's essentially a **conservation law for hardness**: every time you try to dodge a complexity wall with a physical trick, the cost migrates to another conserved quantity (energy, space, time, information, amplitude structure).
- Computational complexity is not just an engineering constraint — it can serve as a genuine **tool of physical explanation.**

7. Further Reading

- **Aaronson, "Why Philosophers Should Care About Computational Complexity"** (2011) — the foundational essay behind the lecture. arXiv: https://arxiv.org/abs/1108.1791. Read this first; everything else in the talk is built on it.
- **Aaronson, PHYS771 Lecture 5 — "Paleocomplexity"** — https://www.scottaaronson.com/democritus/lec5.html. Free online; covers the pre-complexity-theory landscape and why complexity classes matter.
- **Aaronson's blog, *Shtetl-Optimized*** — https://scottaaronson.blog/. Long-form commentary on all the talk's themes (quantum computing hype, P vs NP, Many-Worlds, recent papers).
- **Aaronson's talks archive** — https://www.scottaaronson.com/talks/. Slide decks and venue notes for specific talks.
- **Ewin Tang's dequantization program** — Tang (UW PhD, now Princeton) showed that a celebrated quantum ML recommendation-systems algorithm could be matched classically using analogous sampling techniques. That single result launched a whole line of "dequantization" papers. Start with her PhD thesis: https://ewintang.com/assets/tang_thesis.pdf. The foundational result was published as Tang, "A Quantum-Inspired Classical Algorithm for Recommendation Systems" (STOC 2019, arXiv https://arxiv.org/abs/1807.04271).
- **Shor's algorithm** — Peter Shor, "Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer" (SIAM J. Comput., 1997). arXiv: https://arxiv.org/abs/quant-ph/9508027. The original.
- **Grover's algorithm** — Lov Grover, "A Fast Quantum Mechanical Algorithm for Database Search" (STOC 1996). arXiv: https://arxiv.org/abs/quant-ph/9605043. Original √N search.
- **Post-quantum (lattice-based) cryptography** — NIST's PQC standardization: https://csrc.nist.gov/projects/post-quantum-cryptography. The selected standards (ML-KEM / Kyber, ML-DSA / Dilithium, SLH-DSA / SPHINCS+, FN-DSA / Falcon) are the likely successors to RSA and elliptic-curve crypto.
- **Deutsch, "The Beginning of Infinity"** — David Deutsch, 2011. Broader follow-up that extends the epistemic argument beyond physics.
- **Everett, "Relative State Formulation of Quantum Mechanics"** — Hugh Everett III, Rev. Mod. Phys. (1957). The original Many-Worlds paper.
- **Bohmian mechanics primer** — Goldstein, "Bohmian Mechanics," Stanford Encyclopedia of Philosophy: https://plato.stanford.edu/entries/qm-bohm/.
- **Dynamical collapse (GRW)** — Ghirardi, Rimini, Weber (1986). See Bassi & Ghirardi, "Dynamical Reduction Models," Physics Reports (2003). arXiv: https://arxiv.org/abs/quant-ph/0302164.
- **Harlow & Hayden, "Quantum Computation vs. Firewalls"** — Daniel Harlow and Patrick Hayden, JHEP (2013). arXiv: https://arxiv.org/abs/1301.4504. The paper behind Case Study 3 — argues that decoding Hawking radiation is computationally intractable, dissolving the AMPS firewall paradox.
- **Santa Fe Institute work (Kardes & Hartle, referenced in the talk)** — recent SFI research on the complexity–physics interface. Search SFI's working-paper archive: https://www.santafe.edu/research/working-papers.
- **Morvan et al. (Google), "Phase transitions in random circuit sampling"** (2023–2024). arXiv: https://arxiv.org/abs/2304.11119. The ~70-qubit follow-up — this plus incremental generations gets to the ~103-qubit regime Prof. Aaronson cited.
- **Sean Carroll's *Mindscape*, Episode 99: "Scott Aaronson on Complexity, Computation, and Quantum Gravity"** — long-form conversation covering most of the talk's themes at a gentler pace. https://www.preposterousuniverse.com/podcast/2020/06/01/

99-scott-aaronson-on-complexity-computation-and-quantum-gravity/.